

Methods of proof – 5

Non-constructive existence: counting, diagonal, category,
chance

Homework

13 сентября 2026 года

Theoretical minimum. Four machines produce objects without building them. Counting: countably many obstacles cannot exhaust a continuum of candidates. Diagonalisation: from a countable list one constructs an object differing from every entry, and in the moving form one extracts a subsequence good for countably many conditions at once. Category: in a complete space a residual set is non-empty, so “most” functions have a property that no explicit example may show. Chance: if a random object has the property with positive probability, an object with the property exists. The last two disagree on purpose — a residual set can have measure zero — so “typical” means two different things and it is worth saying which one is meant. Non-constructive proofs are complete proofs; the honest addition is a remark on whether an explicit example is known.

1. Prove that a set of pairwise disjoint intervals on the line is countable, that a family of pairwise disjoint sets of positive measure is countable, and that a monotone function has at most countably many discontinuities (compare with Continuous functions – 8).
2. Prove that there is a real number whose decimal expansion contains every finite word, by counting; and prove that almost every number has this property, by the law of large numbers (Probability – 1).
3. Prove that the set of points of $\mathbb{R}$ that are not the sum of two elements of a given set of measure zero is dense, and construct a set of measure zero whose sum with itself is the whole line.

4. Prove that a countable union of nowhere dense subsets of a complete metric space has empty interior, and give an example of a countable union of closed sets with empty interior that covers a complete space — explaining why this does not contradict Baire.
5. Prove that the set of nowhere differentiable functions is residual in $C[0,1]$ and that the set of functions differentiable at some point is meagre; state the corresponding almost sure statement for the Wiener process.
6. Prove that there exists a set $A \subseteq \mathbb{R}$ such that both A and its complement meet every non-empty perfect set (a Bernstein set), using transfinite induction, and deduce that A is not measurable.
7. Prove that for every countable family of sequences tending to infinity there is a sequence growing faster than all of them, and deduce that no countable family of functions dominates all functions $\mathbb{N} \rightarrow \mathbb{N}$.
8. Prove that a graph on n vertices with independence number and clique number both less than $2 \log_2 n$ exists, by the probabilistic method, and explain why no explicit construction of such a graph is known.
9. Prove that there exists a continuous function on $[0,1]$ whose set of local maxima is dense, and one that is monotone on no interval; use category for the first and an explicit series for the second.
10. Compare the two notions of “typical”: construct a residual set of measure zero in $[0,1]$, deduce that “almost every” and “generic” may contradict each other, and give an example of a property true for almost every number and false for a residual set of numbers.